

BAHRIA UNIVERSITY, ISLAMABAD CAMPUS

Department of Computer Science

Software Engineering (SEN 220)

Assignment # 3 — Spring 2026

Smart Attendance System

(Beyond Biometrics)

Instructor: **Aima Zahoor**

Submission Deadline: **10 May 2026**

Total Marks: **20**

Team Size: **4 Students**



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Milestone 1: Requirements

1.1 Problem Statement

Traditional attendance systems rely on manual registers or basic biometric hardware (fingerprint readers, RFID cards). These are prone to **proxy attendance**, **buddy punching**, **hardware failure**, and high maintenance costs. In large universities like Bahria University with hundreds of students per batch, these approaches create administrative bottlenecks and inaccurate records.

The **Smart Attendance System (Beyond Biometrics)** leverages multiple automated channels — facial recognition, GPS geo-fencing, QR code scanning, and Wi-Fi/BLE proximity detection — to capture attendance seamlessly, accurately, and in real time without dedicated hardware.

1.2 Project Scope

IN SCOPE:

- Automated attendance via facial recognition, GPS geo-fencing, QR codes, and BLE/Wi-Fi proximity
- Role-based access: Admin, Instructor, Student
- Real-time dashboards, attendance statistics, and analytics
- Notification system (email, SMS, push) for absentees and low-attendance warnings
- Reporting module: per student / class / department (CSV/PDF export)
- REST API integration with university LMS/ERP
- Mobile app (Android/iOS) and web portal

1.3 Functional Requirements

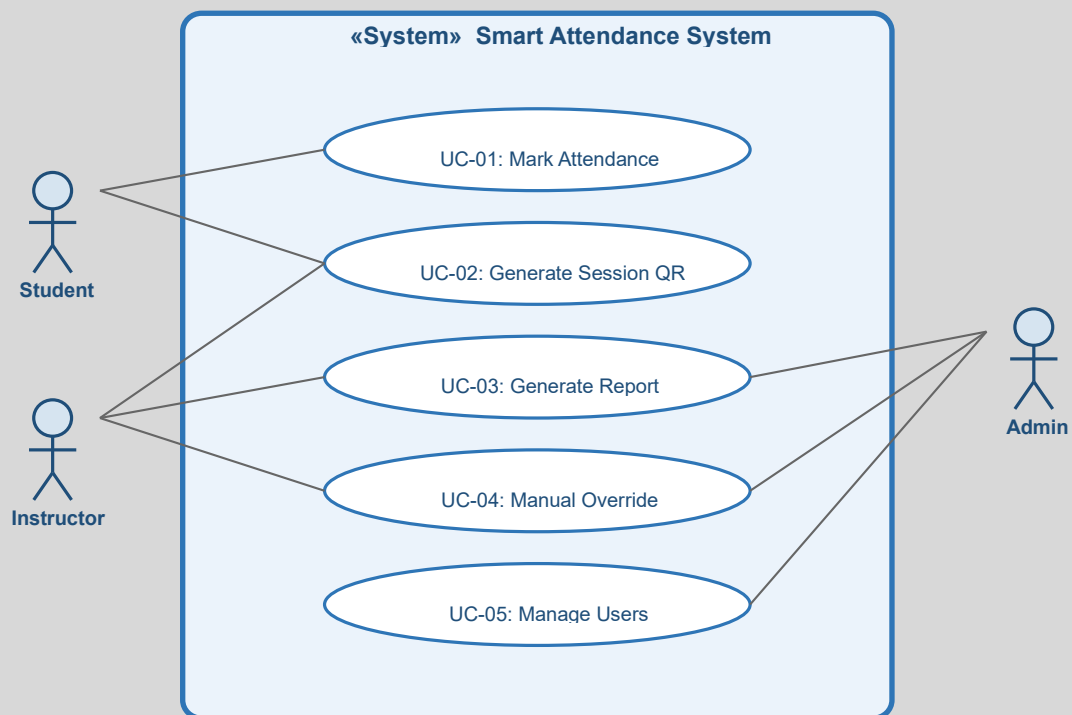
FR #	Requirement	Priority
FR-01	System shall allow students to mark attendance via facial recognition using the device camera.	High
FR-02	System shall verify student physical location via GPS geo-fencing before marking attendance.	High
FR-03	System shall generate time-limited, single-use HMAC QR codes for each class session.	High
FR-04	System shall detect student Wi-Fi/BLE proximity to classroom AP as secondary verification.	Medium
FR-05	System shall allow instructors to manually override attendance records with a mandatory reason.	High
FR-06	System shall send automated alerts when attendance falls below 75%.	High
FR-08	System shall generate daily, weekly, and semester attendance reports.	High
FR-10	System shall maintain an audit trail of all manual corrections with timestamps and actor ID.	High

1.4 Non-Functional Requirements

NFR #	Category	Requirement
NFR-01	Performance	Facial recognition verification shall complete within 3 seconds under normal network.
NFR-02	Scalability	System shall support 500 concurrent active class sessions without degradation.
NFR-03	Availability	System shall maintain 99.5% uptime; maintenance only during off-peak hours.
NFR-04	Security	Face data stored as hashed embeddings (vector), never as raw images.
NFR-05	Security	All API communication uses TLS 1.3; JWT tokens expire within 24 hours.
NFR-06	Usability	Student face enrollment (onboarding) completable in under 5 minutes.
NFR-07	Reliability	Mobile app queues attendance locally if offline and syncs on reconnection.
NFR-08	Compatibility	Mobile app supports Android 10+ and iOS 14+.

1.5 Use Case Diagram

Figure 1.1 — Use Case Diagram: Smart Attendance System



1.6 Use Case Description Tables

UC-01: Mark Attendance

Field	Detail
Use Case ID	UC-01
Use Case Name	Mark Attendance
Actor(s)	Student, Instructor, System
Pre-condition	An active session exists; student is enrolled and authenticated; valid QR code is available.
Main Flow	1. Instructor starts a session and generates QR (UC-02). 2. Student opens the attendance interface. 3. Student scans the session QR code. 4. System validates the QR code and student enrollment. 5. System records attendance as Present with a timestamp. 6. System displays a confirmation to the student.
Alternate Flow	A1: If QR code has expired, system rejects the scan and logs the student as Absent. A2: If student has already marked attendance, system displays "Already marked" message.
Post-condition	Attendance record saved with timestamp; visible to instructor in real time; reflected in reports.
Exception	If QR code is invalid or unrecognized, system rejects the request and displays an error message.

UC-02: Generate Session QR

Field	Detail
Use Case ID	UC-02
Use Case Name	Generate Session QR
Actor(s)	Instructor, Admin, System

Pre-condition	Instructor/Admin is authenticated; a scheduled class session exists in the system.
Main Flow	1. Instructor selects the relevant course and session. 2. Instructor clicks "Generate QR". 3. System creates a unique, time-stamped QR code linked to the session. 4. QR code is displayed on screen for students to scan. 5. System activates an expiry timer for the QR code.
Alternate Flow	A1: If QR expires mid-session, instructor can regenerate a new code for the same session. A2: If no active session is found for the selected slot, system displays an error.
Post-condition	Active QR code linked to the session is available for student scanning; becomes invalid after expiry or session ends.
Exception	If system fails to generate QR due to a server error, instructor is notified and prompted to retry.

UC-03: Generate Report

Field	Detail
Use Case ID	UC-03
Use Case Name	Generate Report
Actor(s)	Instructor, Admin, System
Pre-condition	User is authenticated with Instructor or Admin role; attendance records exist in the system.
Main Flow	1. User navigates to the Reports section. 2. User selects filter criteria (course, student, date range). 3. System retrieves matching attendance records. 4. System displays a summarized report with present/absent counts and percentages. 5. User optionally exports the report as PDF or CSV.
Alternate Flow	A1: If no records match the selected filters, system displays "No records found".

	A2: If export fails, system shows an error message and prompts the user to retry.
Post-condition	Report is displayed and/or downloaded successfully.
Exception	If the system times out while fetching large datasets, user is notified and advised to narrow the filter range.

UC-04: Manual Attendance Override

Field	Detail
Use Case ID	UC-04
Use Case Name	Manual Attendance Override
Actor(s)	Instructor, Admin, System
Pre-condition	Session has ended or is active; instructor has edit rights; target student is enrolled in the course.
Main Flow	1. Instructor searches for the target student. 2. Selects the relevant session record. 3. Changes attendance status (Present / Absent / Late / Excused). 4. Provides a mandatory reason for the override. 5. System validates the changes. 6. System saves the updated record with an audit log entry.
Alternate Flow	A1: If instructor lacks permissions, admin approval is required before the override is applied.
Post-condition	Attendance record updated successfully; reason and audit log stored; reports updated automatically.
Exception	Override after 48-hour lock requires admin approval to unlock and edit the record.

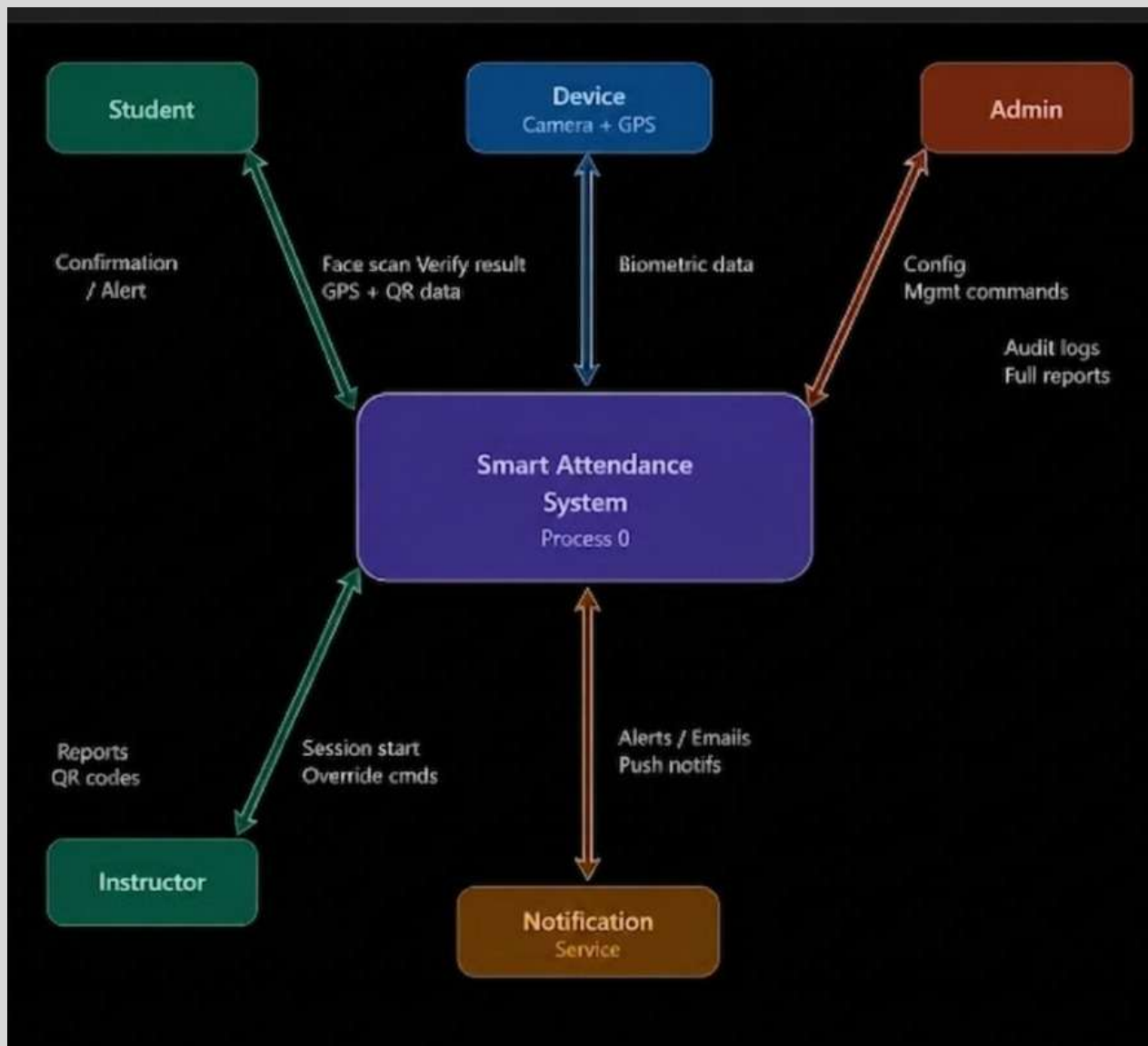
UC-05: Manage Users

Field	Detail
Use Case ID	UC-05
Use Case Name	Manage Users
Actor(s)	Admin, System
Pre-condition	The user must have an Admin role.
Main Flow	1. Admin navigates to User Management. 2. Admin creates, edits, or deactivates a user account. 3. System validates input (unique email, required fields). 4. System saves changes and updates access permissions.
Alternate Flow	A1: If email already exists, system rejects the entry and highlights the conflict. A2: system can save locally.
Post-condition	User accounts are active/inactive as intended
Exception	If bulk import CSV contains invalid rows, system rejects those entries, reports errors per row, and imports the valid ones.

Milestone 2: System Modeling

2.1 Context Diagram (Level 0 DFD)

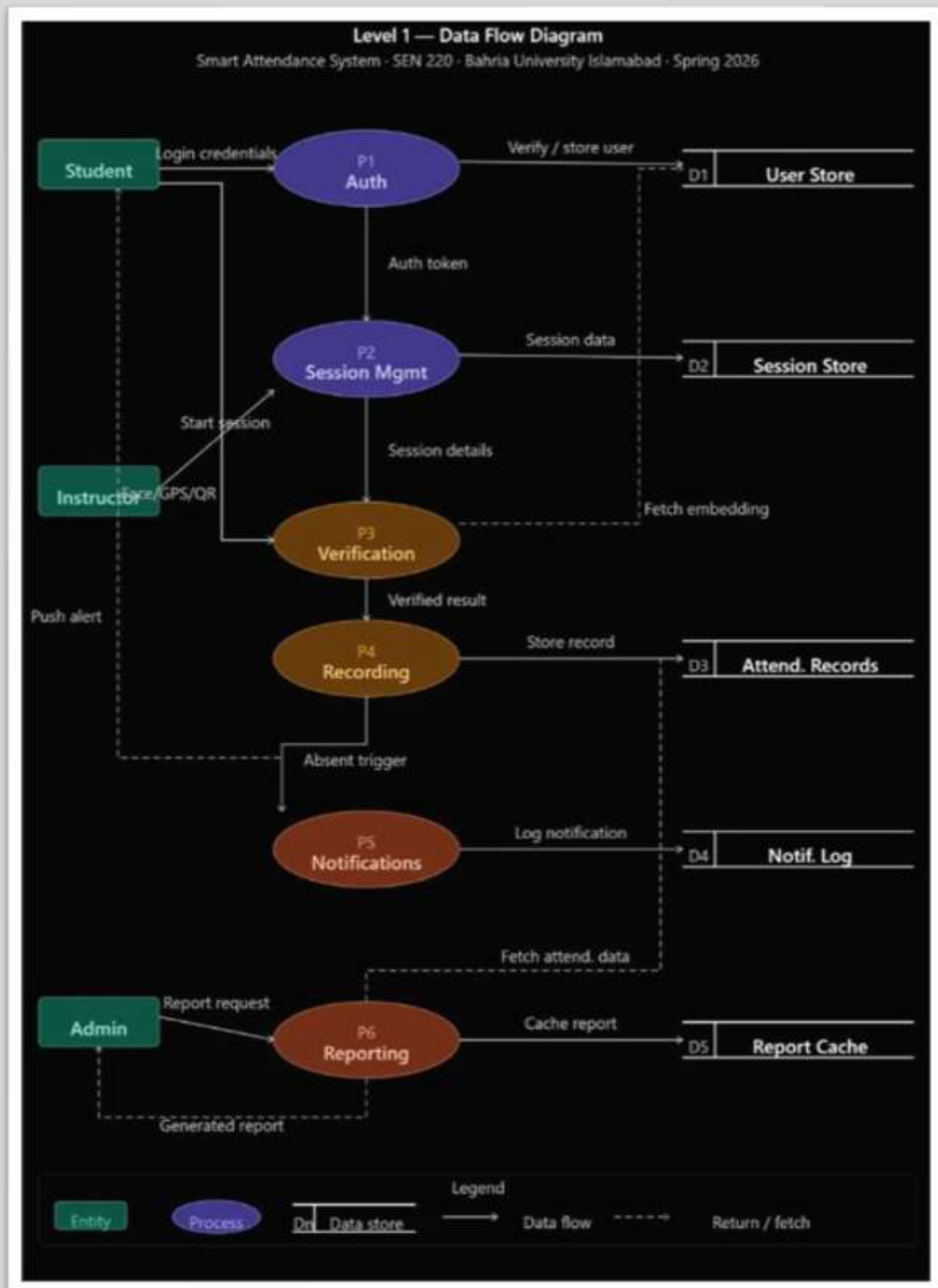
Figure 2.1 — Level 0 Context Diagram



Level 0 — Context Diagram

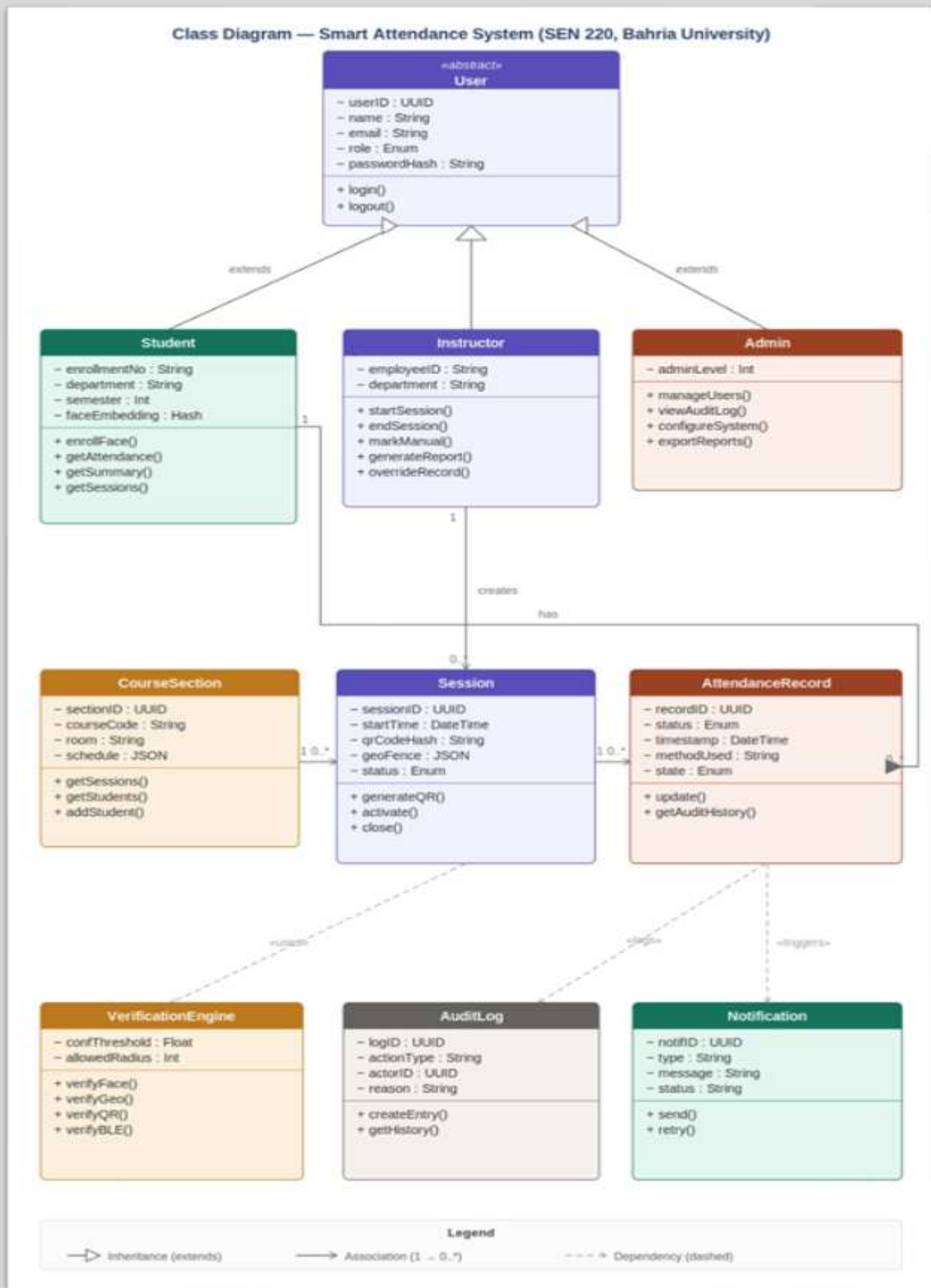
2.2 Level 1 DFD — Major Processes

Figure 2.2 — Level 1 Data Flow Diagram



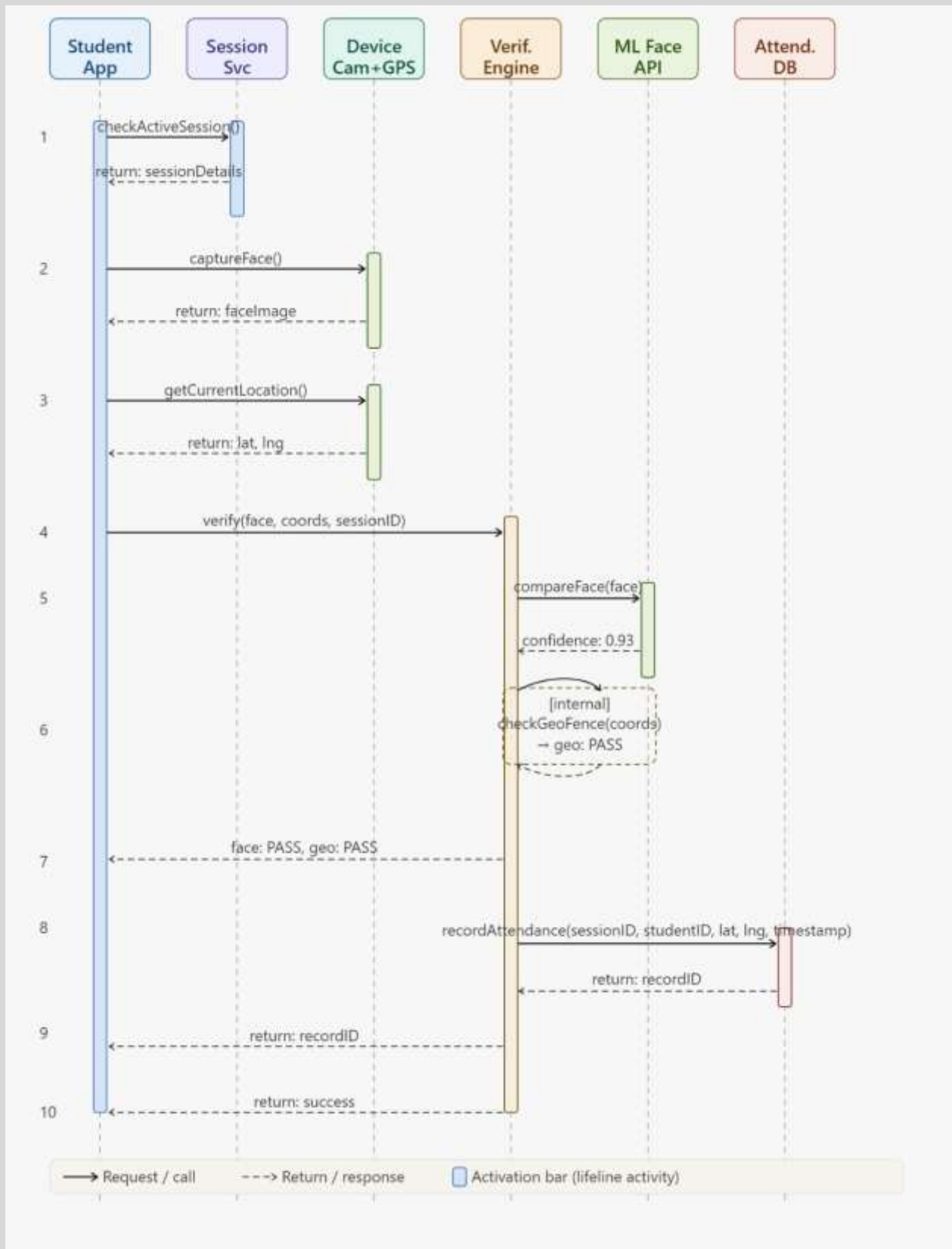
Level 1 — Data Flow Diagram

2.3 Class Diagram



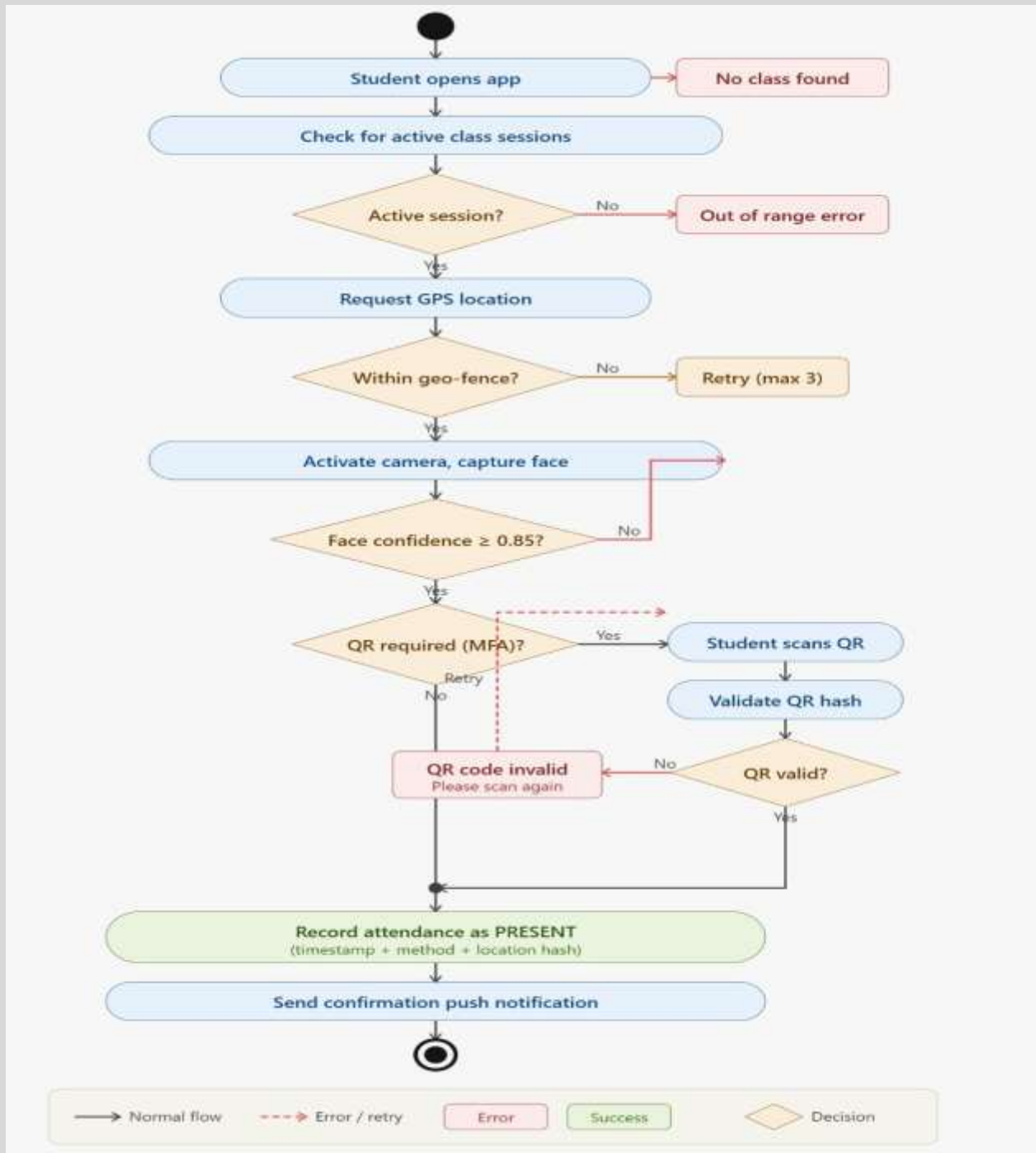
2.4 Sequence Diagram — Mark Attendance

Figure 2.4 — Sequence Diagram: Facial Recognition Attendance Flow



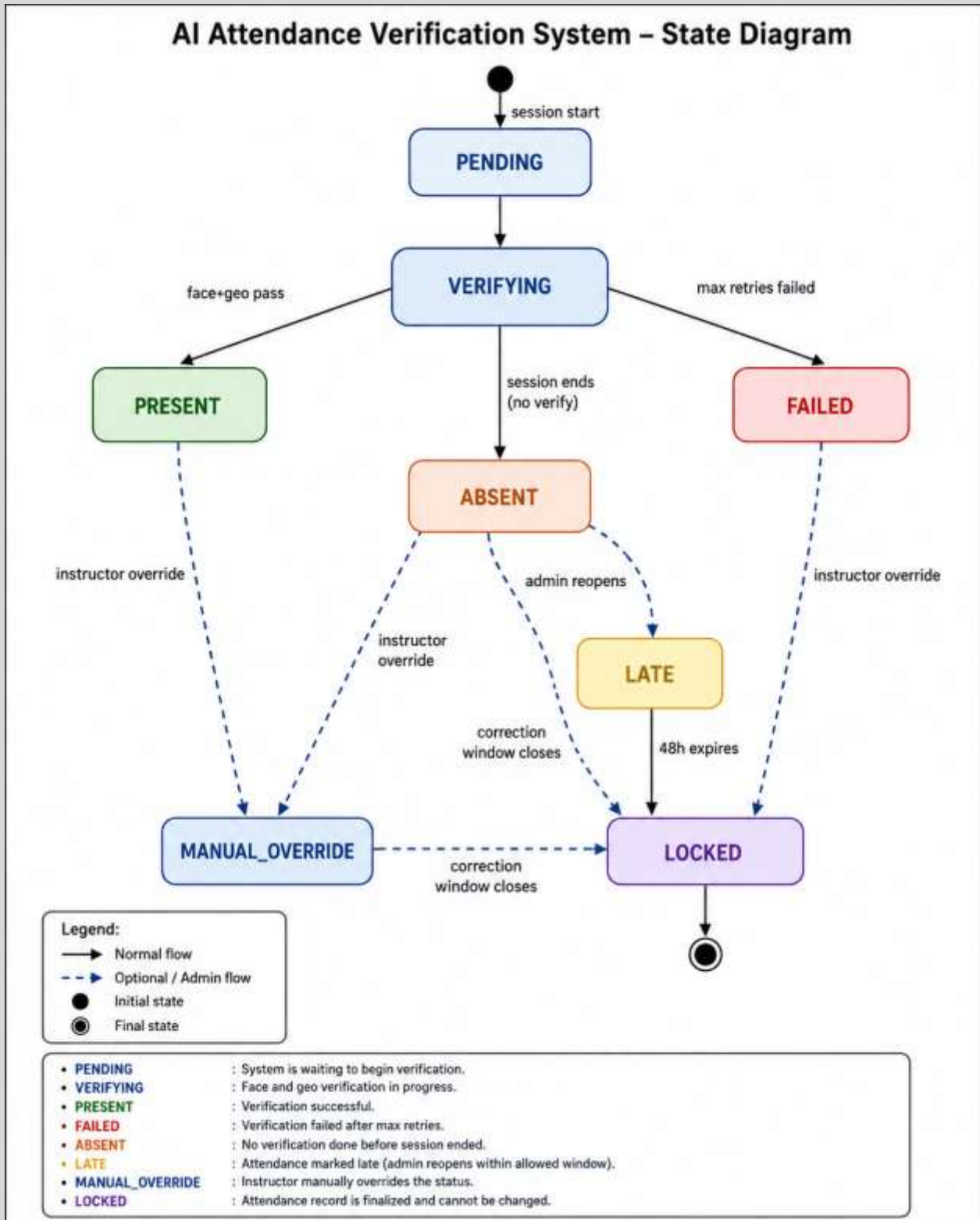
2.5 Activity Diagram — Student Marks Attendance

Figure 2.5 — Activity Diagram: Student Attendance Flow



2.6 State Diagram — AttendanceRecord Lifecycle

Figure 2.6 — State Diagram: AttendanceRecord States

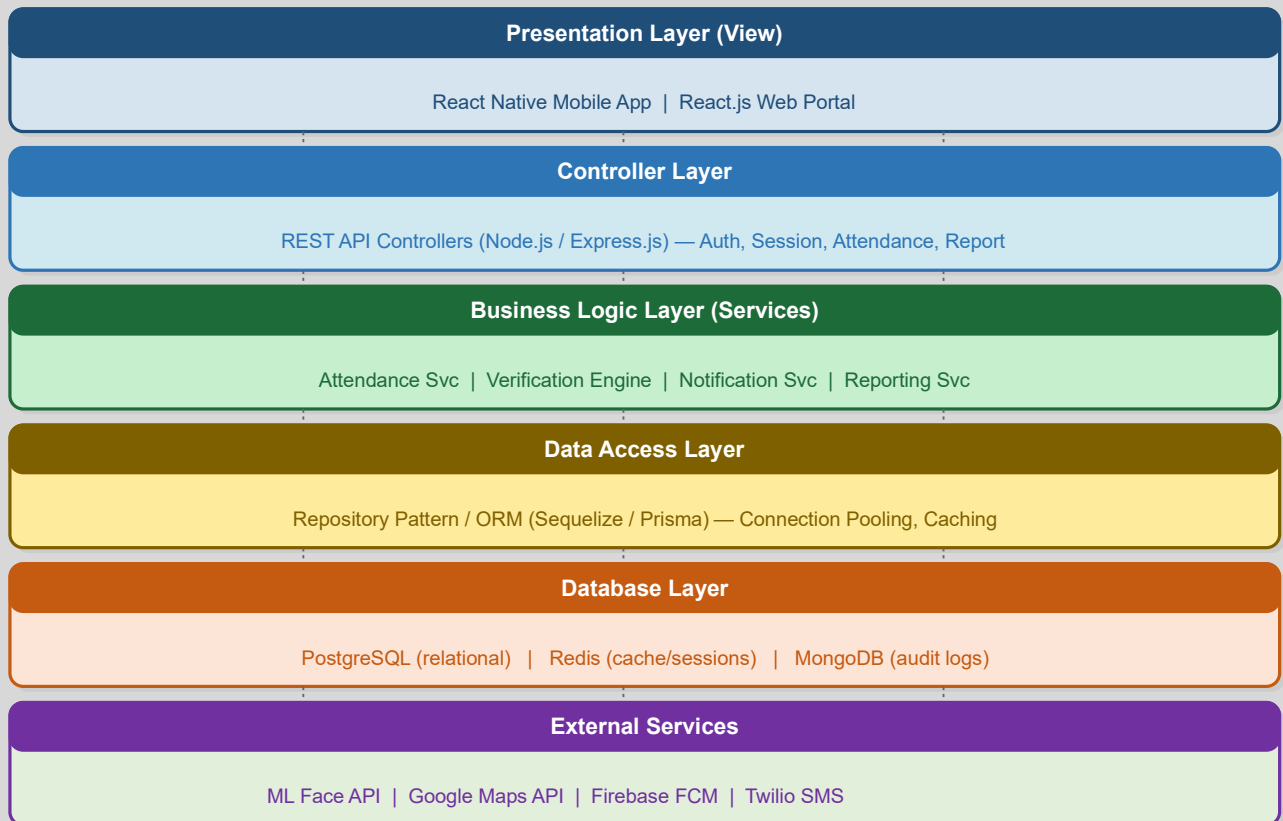


State Diagram — AttendanceRecord Lifecycle

Milestone 3: Design & Architecture

3.1 System Architecture — Layered MVC + Microservices

Figure 3.1 — Layered Architecture Diagram



Layered Architecture Diagram — Smart Attendance System (MVC + Microservices)

3.2 Microservices Breakdown

Service	Port	Responsibility	Tech Stack
Auth Service	3001	User login, JWT issuance, role management, token refresh	Node.js, bcrypt, jsonwebtoken
Session Service	3002	Create/manage class sessions, HMAC-signed QR generation	Node.js, UUID, crypto
Verification Service	3003	Face recognition, geo-fence calc, BLE proximity checks	Python (FastAPI), TensorFlow/FaceNet
Attendance Service	3004	Record CRUD, state machine, override workflow	Node.js, PostgreSQL, Sequelize
Notification Service	3005	Email/SMS/push dispatch with retry logic	Node.js, Firebase FCM, Twilio

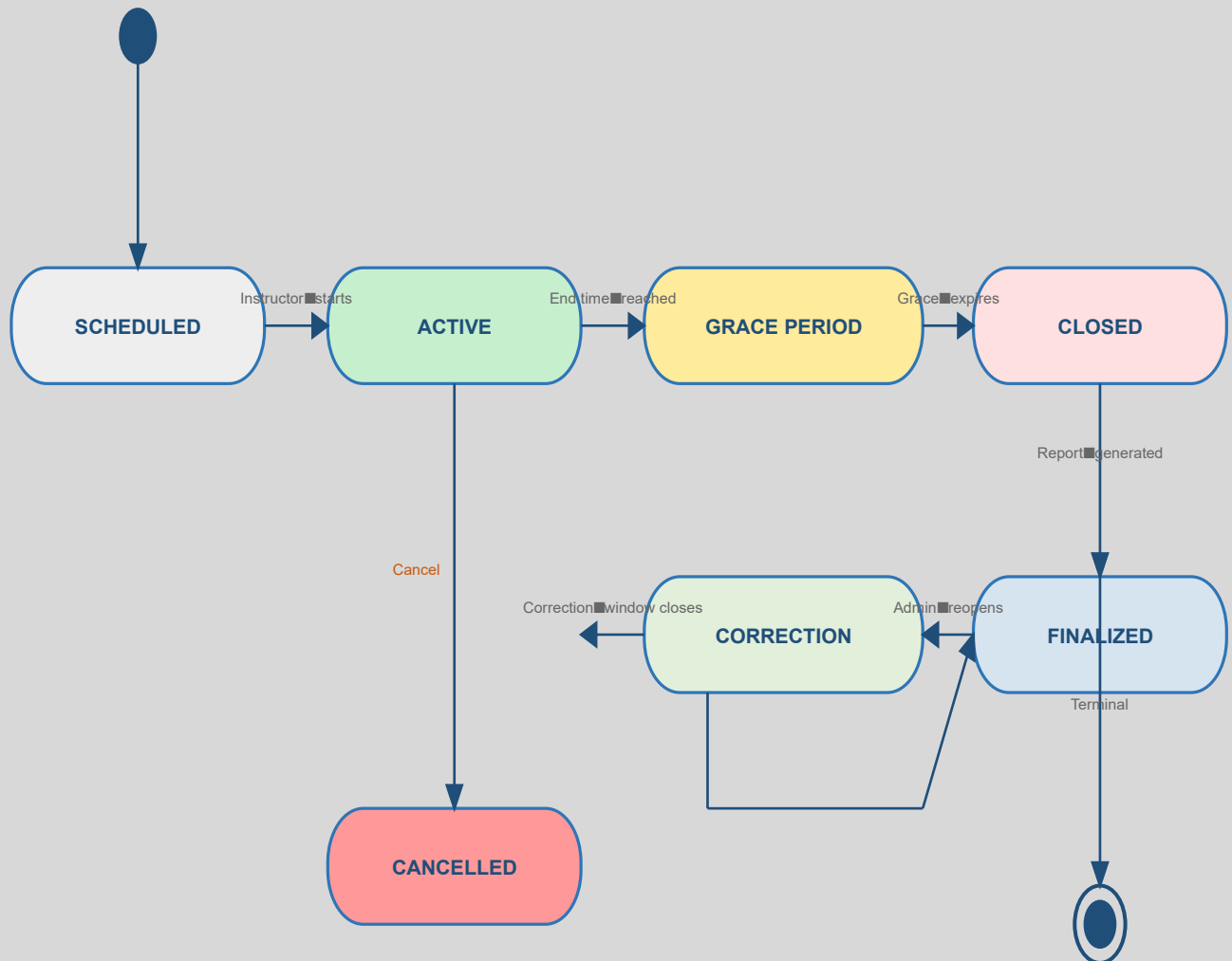
Reporting Service	3006	Aggregate data, generate PDF/CSV exports	Node.js, PDFKit, ExcelJS
API Gateway	443	Routing, rate limiting, SSL, load balancing	Nginx / Kong Gateway

3.3 Database Schema (Key Tables)

Table	Primary Key	Key Columns	Foreign Keys
users	user_id (UUID)	name, email, role, password_hash, created_at	—
students	student_id (UUID)	enrollment_no, department, semester, face_embedding_hash	user_id → users
instructors	instructor_id (UUID)	employee_id, department	user_id → users
courses	course_id (UUID)	course_code, course_name, credit_hours	—
course_sections	section_id (UUID)	section_code, room, schedule_json	course_id, instructor_id
enrollments	enrollment_id (UUID)	enrolled_at, status	student_id, section_id
sessions	session_id (UUID)	start_time, end_time, qr_hash, geo_fence_json, status	section_id
attendance_records	record_id (UUID)	status, method_used, timestamp, location_hash, state	session_id, student_id
audit_logs	log_id (UUID)	action_type, actor_id, reason, old_value, new_value	record_id
notifications	notif_id (UUID)	type, message, channel, status, sent_at	student_id

3.4 State Machine Diagram — Session Lifecycle

Figure 3.2 — State Machine: Session Lifecycle



State Machine Diagram — Session Lifecycle

Milestone 4: Testing & Project Management

4.1 Test Plan

4.1.1 Scope of Testing

- Unit testing — individual service functions and business logic modules
- Integration testing — service-to-service communication and database operations
- System testing — complete end-to-end user flows from mobile app to database
- Performance testing — 500 concurrent users load test
- Security testing — SQL injection, JWT tampering, QR replay attacks

4.2 Sample Test Cases

TC #	Type	Module	Test Scenario	Input	Expected Output	Pass Criteria
TC-01	Unit	Verification	Match known face	Student face image	PASS, conf. 0.93	conf $\geq$ 0.85
TC-02	Unit	Verification	Reject unknown face	Stranger's image	FAIL, conf. 0.41	conf $<$ 0.85
TC-03	Unit	Session Svc	Unique QR per session	sessionID: abc-123	Unique hash, 600s TTL	Hash $\neq$ prev session
TC-04	Unit	Geo-fence	Student within range	Coords $\leq$ 50m from room	Within: TRUE	Result is TRUE
TC-05	Unit	Geo-fence	Student outside range	Coords 300m away	Within: FALSE	Result is FALSE
TC-06	Integration	Attend. + DB	Record saved after verify	Verified payload	DB: status=PRESENT	Record in DB correct
TC-07	Integration	Notif + Attend.	Alert at $<$ 75% attend.	Student attend.=60%	Email+push triggered	Notif record created
TC-08	Integration	Auth + Gateway	JWT needed on routes	No Bearer token	HTTP 401	401 returned
TC-09	System	Override	Instructor edits record	Override + reason	Record+audit+push	All 3 side-effects

4.3 Version Control — GitHub Workflow

Repository: github.com/[team-name]/smart-attendance-system

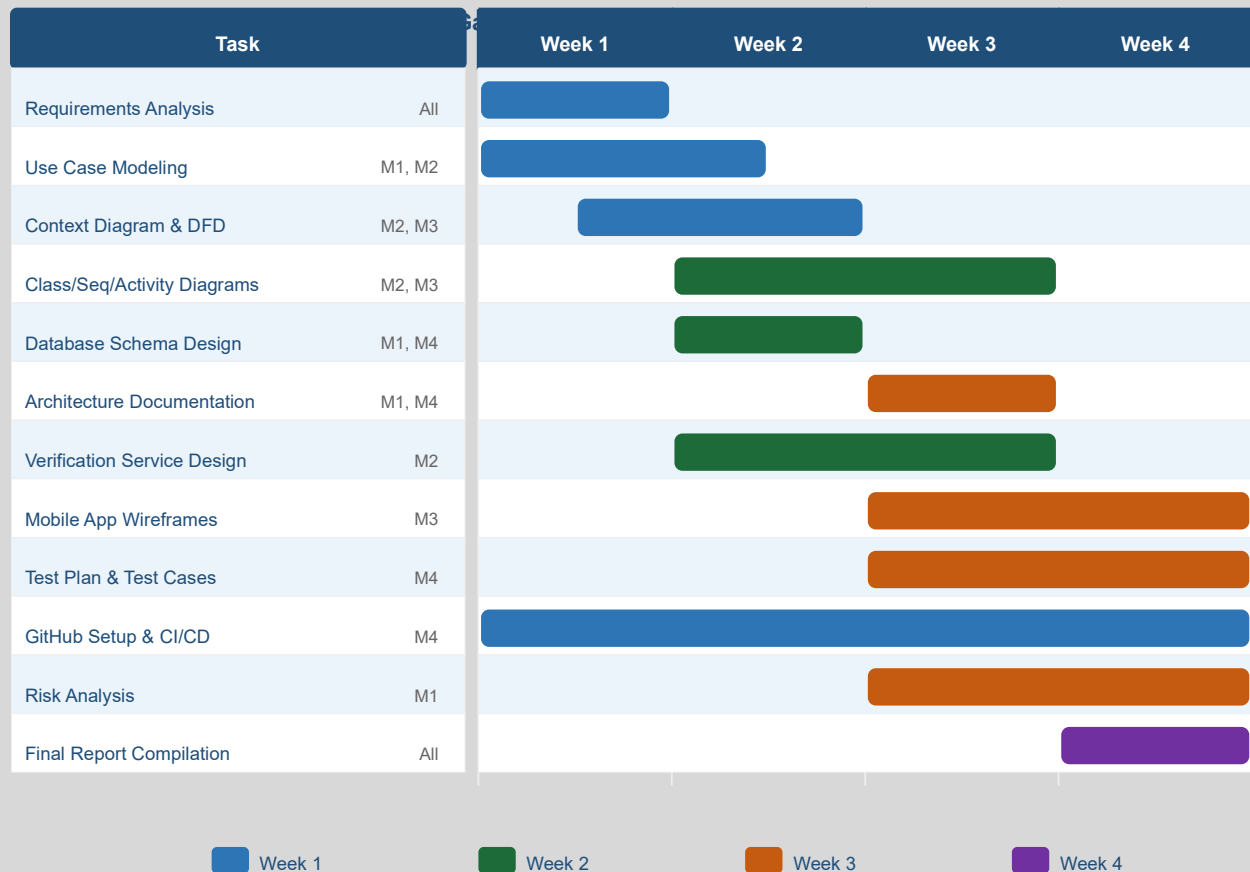
- **Branches:** main (production), develop (integration), feature/* (features), hotfix/*
- **Protection:** PRs require 1 reviewer approval + all CI checks passing before merge to main
- **Convention:** Conventional Commits — feat:, fix:, docs:, test:, chore:

4.4 Team Roles & Responsibilities

Team Member	Role	Primary Responsibilities
ALIZA IBRAR	Project Manager / Backend Lead	Project planning, Gantt chart, risk register, Auth Service and Session Service development
SHAARIFF MUJTABA	ML & Verification Engineer	Face recognition (FaceNet/TensorFlow), Verification Service (Python/FastAPI), geo-fence algorithm
LAIBA TAHIR	Frontend Developer	React Native mobile app, UX/UI design, QR scanning + camera module
MAHAD MALIK	QA Engineer / DevOps	Test case design/execution, CI/CD pipeline, Reporting Service, final documentation

4.5 Project Planning — Gantt Chart

Figure 4.1 — Gantt Chart: 4-Week Project Timeline



4.6 Risk Analysis Register

Risk #	Risk Description	Prob.	Impact	Severity	Mitigation Strategy
R-01	Poor lighting degrades face recognition accuracy	High	High	Critical	Brightness/contrast preprocessing; fallback to QR-only mode in low-light
R-02	Students share QR screenshot remotely	Medium	High	High	HMAC time-limited QR (10 min) + dynamic challenge; combined with geo-fence
R-03	GPS location spoofing by students	Medium	High	High	Cross-validate GPS with BLE/Wi-Fi RSSI; flag impossible location changes
R-04	Server overload during peak attendance	Medium	High	High	Docker horizontal scaling; Nginx load balancer; Redis caching for sessions
R-05	Privacy concerns over facial biometric data	Low	High	High	Store only face embeddings (vectors), not raw images; PDPA-compliant policy
R-06	Device incompatible or app not installed	Medium	Medium	Medium	Web portal fallback; QR-only attendance mode without facial recognition
R-07	Internet outage during classroom session	Low	Medium	Medium	Offline queue in mobile app; sync on reconnect; local BLE session beacon

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